Juliette Bruce's Scholarship, Teaching, Service

Scholarship. My research is at the intersection of algebraic geometry and commutative algebra, which both focus on studying the duality between systems of polynomial equations and the geometry of their solution sets. As described in my future research plan, my work follows three connected threads.

Syzygies. My doctoral work studied the relations, often called syzygies, among systems of polynomial equations that describe higher-dimensional shapes. The equations defining a shape are rarely independent, but instead often satisfy relations, and the relations satisfy further relations, and so on. Homological algebra provides us with tools to organize this cascade of relations and understand what it tells us about the geometry of the underlying shapes. In my work on asymptotic syzygies, I studied how these syzygies change when we describe the same shape using increasingly complicated coordinates. Breakthrough work of Green and of Ein and Lazarsfeld showed that, as the degree of the coordinates goes to infinity, curves have relatively few syzygies (approaching 0% in an appropriate sense), and higher-dimensional varieties have a lot of syzygies (approaching 100% in the limit). My work showed that if one allows more complicated coordinates, where the degrees of different sets of coordinates go to infinity at different rates, one sees intermediate behavior (approaching percentages between 0% and 100%). Such phenomena do not occur for curves and serve to highlight the connections between syzygies and geometry in higher dimensions. This work resulted in publications in the *Journal of Algebra*, *Journal of Commutative Algebra* (showing that the number of syzygies is not always normally distributed), and *Experimental Mathematics* (using high-throughput computing to generate new examples of syzygies for surfaces together with conjectures). Together, this work has laid out a broad foundation for syzygies of higher-dimensional solution sets.

Regularity. Regularity is a numerical measure of complexity that simultaneously captures the complexity of the geometry of a shape and the complexity of the equations defining it. Classically, it is a single number; however, together with Lauren Cranton Heller and Mahrud Sayrafi, I have studied a generalization of this measure in which the single number becomes an infinite staircase-shaped region. We have shown that in this generalized setting, regularity retains some of its nice properties from the classical setting—for example, it can be determined algebraically—but also has a number of pathologies: it is not determined by the syzygies alone, and the staircase can have infinitely many corners. In the latter case, however, my collaborators and I also proved that these pathologies do not arise when studying relatively nice geometric shapes. Using these results, we also proved that as one takes powers of the defining polynomial equations, the regularity staircases exhibit surprising uniformity, in that they translate in a fixed direction at a linear rate in terms of the powers being taken. This work has resulted in one publication in *Advances in Mathematics* and two preprints under review at journals of roughly the same quality. Together with my work on syzygies, this creates a unified research program seeking to extend the classical homological algebra tools developed in the 19th and 20th centuries to study projective spaces into tools adapted to studying more complicated geometric spaces.

Tropical Geometry of Moduli Spaces. Tropical geometry seeks to study moduli spaces—geometric spaces whose points package together families of solutions to polynomial equations—using the combinatorics of polyhedra, graphs, and more complicated objects like matroids and cell complexes. Using these methods with a number of collaborators, I have studied the geometry of moduli spaces that arise naturally in number theory: the moduli space of abelian varieties and Shimura varieties. In doing so, we have constructed new cohomology classes, which can be thought of as measuring the number of holes in a shape, and developed new combinatorial tools for understanding these spaces. For the moduli space of abelian varieties, these classes answered a decade-old question of Grushevsky about the existence of cohomology classes in odd degrees. Further, our work highlighted connections between algebraic geometry and a seemingly unrelated area of mathematics, the study of arithmetic groups of matrices. The resulting connections have attracted substantial interest in recent years. This work was highlighted in a plenary lecture by Melody Chan at the 2025 Summer Research Institute, a conference held every ten years that highlights some of the best work from the preceding decade. These efforts have resulted in one publication in *Geometry & Topology* and three preprints under review at journals.

Dissemination. Since joining Dartmouth in fall 2024, I have posted five new preprints, all of which are under review at peer-reviewed journals. My paper characterizing multigraded regularity was accepted in *Advances in Mathematics*. Since June 2024, I have given 12 invited research talks at seminars and

conferences. I have attended six invitation-only research workshops since 2024, including two at Oberwolfach and others at the Banff International Research Station, the Fields Institute, and the American Institute of Mathematics.

Teaching and Mentoring. My first experiences teaching were as an undergraduate course assistant in inquiry-based courses at the University of Michigan. In these courses, the students worked together to build and discover mathematics for themselves through a carefully curated mix of group work, lectures, and carefully chosen problems. That experience still shapes much of my teaching. I want students to have opportunities to discover and work out an idea, explain it to someone else, and develop their own questions.

At Dartmouth, I have taught introductory calculus, undergraduate abstract algebra, and an array of graduate courses. At the undergraduate level, most of my teaching focus has been on refining and improving Math 8: Calculus of One and Several Variables. This course serves as the first mathematics course many Dartmouth students take and, at least according to the students in my first class, has a somewhat intimidating reputation. I have spent significant time making this course a point of support for students, where they are challenged to meet high academic standards in a supportive environment.

The advanced courses I have taught (Math 101, 121, 125) are where my teaching and research meet most directly. Math 125 was a topics course using toric varieties as a touchstone to provide students with a grounded introduction to algebraic geometry. Math 121 was a commutative algebra course, which I designed around a series of sixteen inquiry-based worksheets that build on the work of Karen Smith. These worksheets are freely available on my website, and I hope to slowly develop them into an open-source textbook. Additionally, I have taught Math 150, a graduate reading course, twice; in fall 2026, I led a group of approximately 10 students studying positivity in algebraic geometry, and in winter 2026, I led a group of approximately 15 undergraduate and graduate students learning about one of the jewels of 20th-century mathematics: the Weil conjectures.

Currently, I am advising two Ph.D. students: Will DeGroot (4th year) and Jacob Lehmann Duke (3rd year). Their projects connect with my own interests in computation and the complexity of equations on toric varieties. I expect that within the year I will become Alex Moon's (2nd year) Ph.D. advisor, as I have been helping them prepare for their advancement exam and advising them on a small research project. Since summer 2026, I have mentored Amy Q. Li, a new postdoc in the mathematics department. Beyond Dartmouth, in summer 2026, I led three undergraduates in a summer research program through *Count Me In* held at the University of Edinburgh. All three of these students are now in the process of applying to graduate school. In fall 2026, I am one of two lecturers for the Preliminary Arizona Winter School, a nine-week online course introducing approximately 300 advanced undergraduate students and early graduate students from around the world to recent advancements in computational algebraic geometry.

Service and Community. *At Dartmouth.* In fall 2025, I founded *Unwritten Rules*, a graduate seminar that seeks to make the hidden curriculum and expectations of an academic career explicit. Since its founding, this seminar has met over dinner 2–3 times a term to discuss things like attending conferences, applying for funding, and submitting papers to journals. A side effect of this has been an increased sense of community among the graduate students. I serve as the faculty co-advisor for Dartmouth's student chapter of the Association for Women in Mathematics. In November 2024, I organized the Algebraic Geometry Northeastern Series (AGNES) and BATMOBILE conferences at Dartmouth, which brought over 150 mathematicians to Dartmouth to discuss recent breakthroughs in algebraic geometry. Currently, I am in the early planning stages of organizing *GEMS of Commutative Algebra*, a conference building research networks for early-career researchers in commutative algebra, which I hope to hold at Dartmouth in 2028.

In the profession. I was the inaugural president of *Spectra, the Association for LGBTQ+ Mathematicians*, led the adoption of bylaws and the creation of an invited lecture series at the Joint Mathematics Meetings, and raised over $20,000 in seed funding. I am currently serving on the board of trustees for *Spectra*, working to guide the organization toward long-term organizational stability. From 2023 to 2025, I served on SLMath's Committee on Women in Mathematics. Currently, I serve on the American Mathematical Society's Committee on Engagement, Participation, and Advancement and the Committee on AMS-Simons Travel Grants. I have reviewed grants for the National Science Foundation and refereed for more than a dozen different journals. In my career, I have organized 18 conferences and workshops that have brought together over 1,000 mathematicians. With a co-organizer, I have submitted a proposal to the Fields Institute to organize a thematic semester in combinatorial algebraic geometry.